

HADOOP INSTALL

COMMANDS

HADOOP VERSION – 2.9.0

JAVA – 8 (1.8.0)

WORKS ON ANY UBUNTU VERSION NO HDUSER REQUIRED

Installing and moving (make sure the java version is same as given below)

```
sudo apt install openjdk-8-jdk -y
```

```
java -version
```

```
sudo apt install ssh -y
```

```
ssh-keygen -t rsa -P ""
```

```
cat ~/.ssh/id_rsa.pub >> ~/.ssh/authorized_keys
```

```
chmod 600 ~/.ssh/authorized_keys
```

```
ssh localhost
```

```
tar -xzf hadoop-2.9.0.tar.gz
```

```
sudo mv hadoop-2.9.0 /usr/local/hadoop
```

Set Environment Variables

```
nano ~/.bashrc ### go to end and paste it
```

```
export JAVA_HOME=/usr/lib/jvm/java-8-openjdk-amd64
```

```
export HADOOP_HOME=/usr/local/hadoop
```

```
export PATH=$PATH:$HADOOP_HOME/bin
```

```
export PATH=$PATH:$HADOOP_HOME/sbin
```

```
export HADOOP_MAPRED_HOME=$HADOOP_HOME
```

```
export HADOOP_COMMON_HOME=$HADOOP_HOME
export HADOOP_HDFS_HOME=$HADOOP_HOME
export YARN_HOME=$HADOOP_HOME
export HADOOP_COMMON_LIB_NATIVE_DIR=$HADOOP_HOME/lib/native
export HADOOP_OPTS="-Djava.library.path=$HADOOP_HOME/lib"
export HADOOP_CONF_DIR=/usr/local/hadoop/etc/hadoop

source ~/.bashrc
```

Step 3: Configure Hadoop

```
cd $HADOOP_HOME/etc/hadoop
```

```
nano core-site.xml
```

```
<configuration>
  <property>
    <name>fs.defaultFS</name>
    <value>hdfs://localhost:9000</value>
  </property>
</configuration>
```

```
nano hdfs-site.xml
```

```
<configuration>
  <property>
    <name>dfs.replication</name>
    <value>1</value>
  </property>
  <property>
```

```
<name>dfs.namenode.name.dir</name>

<value>file:///usr/local/hadoop/hdfs/namenode</value>

</property>

<property>

  <name>dfs.datanode.data.dir</name>

  <value>file:///usr/local/hadoop/hdfs/datanode</value>

</property>

<property>

  <name>dfs.namenode.rpc-address</name>

  <value>localhost:9000</value>

</property>

<property>

  <name>dfs.namenode.http-address</name>

  <value>localhost:9870</value>

</property>

</configuration>
```

cp mapred-site.xml.template mapred-site.xml

nano mapred-site.xml

```
<configuration>

  <property>

    <name>mapreduce.framework.name</name>

    <value>yarn</value>

  </property>

</configuration>
```

nano yarn-site.xml

```
<configuration>
  <property>
    <name>yarn.resourcemanager.address</name>
    <value>localhost:8032</value>
  </property>
  <property>
    <name>yarn.nodemanager.aux-services</name>
    <value>mapreduce_shuffle</value>
  </property>
</configuration>
```

nano hadoop-env.sh

```
export JAVA_HOME=/usr/lib/jvm/java-8-openjdk-amd64
```

hdfs namenode -format

start-dfs.sh

start-yarn.sh

go to <http://localhost:9870> for Hadoop dashboard and <http://localhost:8088> for manager

MAP REDUCE

COMMANDS

Create a folder called analyzelogs in the home and paste following files in it:

- 1) Link CLASSPATH & MANIFEST
- 2) Link log file
- 3) Mapper
- 4) Reducer
- 5) Driver

These files can be downloaded from the url -

<https://www.youtube.com/watch?v=gXQT7vHB59Q>

```
cd ~/analyzelogs
```

convert the accesslog.txt to csv by opening it in libre calc and setting rows 2, separated by other –

#####

```
ls
```

```
export CLASSPATH="$HADOOP_HOME/share/hadoop/mapreduce/hadoop-mapreduce-client-core-2.9.0.jar:$HADOOP_HOME/share/hadoop/mapreduce/hadoop-mapreduce-client-common-2.9.0.jar:$HADOOP_HOME/share/hadoop/common/hadoop-common-2.9.0.jar:~/analyzelogs/SalesCountry/*:$HADOOP_HOME/lib/*"
```

```
javac -d . SalesMapper.java SalesCountryReducer.java SalesCountryDriver.java
```

```
nano Manifest.txt
```

paste following in the Manifest.txt

```
Main-Class: SalesCountry.SalesCountryDriver
```

#####

```
jar -cfm analyzelogs.jar Manifest.txt SalesCountry/*.class
```

```
start-dfs.sh
```

```
start-yarn.sh
```

```
mkdir ~/input2000
```

```
cp access_log_short.csv ~/input2000/
```

```
$HADOOP_HOME/bin/hdfs dfs -put ~/input2000 /
```

```
$HADOOP_HOME/bin/hadoop jar analyzelogs.jar /input2000 /output2000
```

```
$HADOOP_HOME/bin/hdfs dfs -cat /output2000/part-00000
```

```
jar -cfm analyzelogs.jar Manifest.txt SalesCountry/*.class
```

```
cp access_log_short.csv ~/input2000/
```

```
$HADOOP_HOME/bin/hdfs dfs -put ~/input2000 /
```

```
$HADOOP_HOME/bin/hadoop jar analyzelogs.jar /input2000 /output2000
```

HIVE INSTALL

COMMANDS

HIVE VERSION – 1.2.2

JAVA – 8 (1.8.0)

```
sudo cp apache-hive-1.2.2-bin.tar.gz /usr/local/
```

```
nano ~/.bashrc ### go to end and paste it
```

```
    export HIVE_HOME=/usr/local/hive
```

```
source ~/.bashrc
```

```
cd /usr/local
```

```
sudo tar -xvzf apache-hive-1.2.2-bin.tar.gz
```

```
sudo mv apache-hive-1.2.2-bin hive
```

```
start-dfs.sh
```

```
start-yarn.sh
```

```
cd hadoop/bin/
```

```
hdfs dfs -mkdir -p /user/hive/warehouse
```

```
hdfs dfs -mkdir -p /tmp/hive
```

```
hdfs dfs -chmod 777 /tmp
```

```
hdfs dfs -chmod 777 /user/hive/warehouse
```

```
hdfs dfs -chmod 777 /tmp/hive
```

```
cd /usr/local/hive/bin/
```

```
sudo chmod -R 755 /usr/local/hive
```

```
sudo chown -R omkar:omkar /usr/local/hive
```

```
./schematool -initSchema -dbType derby
```

```
./hive
```